Mohammed Hamdy

Artificial Intelligence Engineer

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Summary

AI Engineer with hands-on production experience building LLM agents, RAG systems, and multimodal AI platforms and running them on cloud-native infrastructure. Built LangGraph multi-agent assistants with short- and long-term memory, human-in-the-loop approvals, and secure SQL tools, agentic Text-to-SQL with validation and self-repair, hybrid RAG (BM25 + embeddings + reranking) with offline evaluation, LLM guardrails (PII redaction, prompt-injection defense, output validation), and vision-LLM pipelines for bilingual Arabic/English compliance analysis and image generation. Migrated a multi-product AI platform from Docker Compose to Kubernetes with ArgoCD GitOps, Helm, Sealed Secrets, and full observability (Prometheus, Grafana, Loki, Langfuse).

Skills

LLM & Agents
LangGraph, LangChain, Tool calling, Multi-agent systems, Human-in-the-loop, Structured outputs, Prompt engineering, Context caching, Text-to-SQL

RAG
Hybrid search (BM25 + embeddings), Reranking, RRF, FAISS, ChromaDB, Retrieval evaluation (hit@k, MRR)

LLM Safety
Guardrails, PII protection, Prompt-injection defense, Output validation

Serving
vLLM, HuggingFace, Vertex AI, AWS Bedrock

Backend
Python, FastAPI, PostgreSQL, Celery, RabbitMQ, Redis, Async I/O, pytest

MLOps & DevOps
Docker, Kubernetes, Helm, ArgoCD (GitOps), Sealed Secrets, CI/CD, GitHub Actions

Observability
Langfuse, Prometheus, Grafana, Loki, LLM cost tracking

Cloud
Azure, AWS, GCP, Alibaba Cloud

Certifications

- **Kubernetes in Production**, *Cloud Native Base Camp* · View certificate
- **Git for Production Teams**, *Cloud Native Base Camp* · View certificate
- **Cloud-Native Learning Track**, *Cloud Native Base Camp* · Linux, containers, DevOps, cloud, and site reliability engineering · View profile

Professional Experience

AI Engineer, *VEEM Solutions* Aug 2024 – Present

Shrwd — AI Brand-Intelligence & Social-Listening Platform

- Designed and built 8 FastAPI microservices (brand analysis, media monitoring, strategy generation, creative generation, conversational agent), each Dockerized and shipped through GitHub Actions CI/CD to staging and production
- Built **Shrwdy**, a LangGraph multi-agent assistant with three tool-calling ReAct agents (brand analytics, strategy, creative editing), SSE streaming, and human-in-the-loop tool approvals; Postgres-backed memory works across pods, with rolling summarization to cut per-turn token cost and long-term carry-over between sessions
- Secured LLM data access: the agent reads data only through read-only SQL tools over row-level-scoped views, with the user identity injected server-side so the model can never query another user's data; tool errors return the valid values so the model corrects itself
- Built hybrid RAG (BM25 + dense embeddings, RRF fusion, local cross-encoder reranker) with graceful per-stage fallbacks and an offline evaluation harness (hit@k, MRR, optional LLM-judge metrics)
- Built LLM guardrails without extra LLM calls: PII redaction with reversible placeholders, prompt-injection isolation of third-party content, and validators (schema, allowed numbers, language, length, platform rules) that reject invented numbers and retry the model with the errors
- Built a multimodal creative pipeline: brief → platform-specific posts → Gemini image generation → vision-model QA loop that catches painted text, fake logos, and defects, then repairs or regenerates; brand logo, Arabic-shaped typography, and QR codes are composed in code, and every design is editable with full version history
- Built web crawling and extraction (httpx with a Playwright fallback for bot-protected sites, plus search fallbacks) for brand assets, brand profiles, and media coverage, using LLM query planning with deterministic relevance scoring
- LLMOps: per-request cost accounting (tokens, image, search, speech), Gemini context caching, and cross-pod cancellation via Postgres NOTIFY that stops work cleanly without losing its cost

Tech Stack: Python · FastAPI · LangGraph · LangChain · Gemini / Vertex AI · PostgreSQL · Azure Blob Storage · Playwright · Docker · GitHub Actions · pytest

SyntraAI (GlassHub) — AI Compliance Analysis & Agentic Analytics Platform

- Built a FastAPI service for multimodal compliance analysis that reviews evidence files (PDF, image, Excel) against a control's requirements using Qwen vision/text LLMs, returning structured bilingual (English/Arabic) assessments plus PDF, Excel, and Markdown reports stored on Alibaba Cloud OSS
- Built an agentic Text-to-SQL system on LangGraph as a 7-node graph (Router → Planner → SQL Generator → Validator → Executor → Repair → Answer), with a safety validator re-checking every repaired query, a bounded repair budget, and graceful decline of out-of-scope questions
- Built a "Master Agent" for comparing compliance frameworks: a 3-stage exact → name → semantic (LLM) matcher for one organization, and a multi-organization harmonization pipeline (deterministic matching, embedding similarity, LLM consolidation and validation) that preserves lineage, with tunable concurrency and token budgets
- Added an agentic heatmap assistant (Compliance Exposure Map) that classifies questions as data vs. chat and generates SQL scoped to the dashboard's active filters
- Cut repeat vision-LLM costs with a PostgreSQL-backed page/document cache that falls back to a file cache
- Built in observability: Langfuse tracing on every endpoint, structured Loki logging with X-User-ID / X-Request-ID request tracing, and Loki recording rules feeding Prometheus metrics into incident and SLA Grafana dashboards

Tech Stack: Python · FastAPI · LangGraph · LangChain · Qwen (text + vision) · Sentence-Transformers · FAISS · PostgreSQL · SQLAlchemy · PyMuPDF · Pandas · Alibaba Cloud OSS · Langfuse · Loki · Prometheus · Grafana

CST (GlassHub) — AI Regulatory Compliance Agents for Regulators & Enterprises

- Built a FastAPI platform with separate Regulator and Enterprise API domains (licenses & organizations), each with admin, data, main-process, and chat routers
- Built an LLM document-understanding pipeline that extracts regulatory terms from Arabic/English Word and PDF regulations and generates structured audit controls (audit and clause-level instructions, language detection), with async LLM calls under a bounded concurrency limit
- Built RAG over regulatory knowledge: chunking → multilingual embeddings → FAISS, with hybrid retrieval (semantic + BM25 via EnsembleRetriever) and metadata-filtered search by company and criteria
- Built vision-based compliance reporting: evidence PDFs and images are rendered and analyzed by a Qwen vision-language model that returns compliance status, flags, a summary, and a human-review flag in Arabic or English
- Built 3 domain chatbots (regulatory RAG chat, per-term chat, non-compliance remediation chat), each with its own context filters
- Instrumented with Prometheus (custom per-endpoint metrics) and shipped via Docker, NGINX path-based routing, and GitHub Actions staging/production pipelines

Tech Stack: Python · FastAPI · LangChain · Qwen3 / Qwen-VL · FAISS · BM25 · Sentence-Transformers · PyMuPDF · Tesseract OCR · Alibaba Cloud OSS · Prometheus · Docker · NGINX

GlassHub Platform — Kubernetes GitOps Migration, LLM Gateway & Model Serving

- Migrated 5 Docker Compose projects (~17 containers) from a single VM to K3s on Azure using ArgoCD's app-of-apps pattern: one root Application bootstraps 14 child Applications with automated sync, prune, and self-heal
- Wrote Helm charts for every service (Deployments with HPA, Ingress, and probes; StatefulSet + PVC for the PostgreSQL cache) with per-environment values for staging and production namespaces
- Built a build-once, promote-to-prod CI/CD pipeline in GitHub Actions: production verifies the staging-built image exists and bumps its tag in the GitOps repo with yq, and ArgoCD rolls it out, with no manual helm or kubectl steps
- Secured the platform with Bitnami Sealed Secrets and default-deny NetworkPolicies (only the AI apps can reach the LLM gateway; only SyntraAI can reach its Postgres cache), behind ingress-nginx path-based routing on a single public endpoint
- Deployed observability as code: kube-prometheus-stack (Prometheus, Grafana, Alertmanager), Loki + Promtail, and self-hosted Langfuse (Postgres, Valkey, ClickHouse, cert-manager) using multi-source ArgoCD Applications and sync waves
- Built **LLM Cipher**, a centralized LLM gateway (FastAPI + Gunicorn) with chat/vision endpoints and API-key auth, so models and providers can change without touching client apps; self-hosted Qwen2.5-VL and Qwen3 on GPU with vLLM + FlashAttention, with bf16, GPU-memory, and context-length tuning
- Debugged live-cluster issues and fixed each in code (CRD annotation-size sync failures via Server-Side Apply, Grafana OOMKills, a datasource CrashLoopBackOff, a Promtail DNS mismatch), and documented operations in 7 runbooks

Tech Stack: Kubernetes (K3s) · Azure · ArgoCD · Helm · Sealed Secrets · GitHub Actions · ingress-nginx · Prometheus · Grafana · Alertmanager · Loki · Langfuse · ClickHouse · vLLM · FlashAttention · Docker

vminds Chatter — Multimodal AI Orchestration Platform

- Designed and built a production-ready AI orchestration backend unifying chat, image generation, audio synthesis, document understanding, and spreadsheet analysis into a single API-driven service
- Architected a modular routing system mapping user intent to specialized AI capabilities across multiple LLM providers (Gemini / AWS Bedrock)
- Built reusable model wrappers to abstract provider-specific logic, enabling seamless switching across cloud AI services
- Implemented environment-aware architecture with isolated staging and production routers for safe experimentation and controlled deployments
- Introduced async file handling and temporary caching to improve document processing throughput; developed robust JSON sanitization for unreliable LLM outputs
- Implemented token usage tracking and cost-aware execution strategies for efficient AI operations

Tech Stack: Python · FastAPI · Pydantic · LangChain · AWS Bedrock · Gemini (Vertex AI) · Docker Compose · NGINX · Cloud virtual machines

vminds Solo — AI Media Processing Microservices Platform

- Built a production-ready AI microservices platform for media transformation and speech processing (background removal, TTS, transcription, image enhancement)
- Architected an async task processing system using Celery and RabbitMQ, decoupling the API layer from worker execution to handle long-running ML workloads without blocking requests
- Deployed and managed containerized services on Kubernetes — configuring deployments, services, secrets, and config maps across multiple microservices
- Implemented task-level routing with dedicated Celery queues per job type and tuned worker lifecycle parameters (concurrency, prefetching, late ACKs) for optimal fault tolerance
- Integrated Google Cloud AI services for TTS and image processing alongside custom pipelines for media manipulation

Tech Stack: Python · FastAPI · Celery · RabbitMQ · Docker · Kubernetes · Google Cloud AI

Education

B.Sc. Computer Science & Information Systems, *City of Culture and Science University* — 2018 – 2022
October, Egypt